Alberta's Industrial Emissions, 2009 to 2024

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Summary

Alberta's large industrial facilities emitted 141.6 Mt CO2e in 2024, 20% more than in 2009. That total hides two opposite trends. Oil, gas and mining emissions nearly doubled, from 50.1 to 91.3 Mt, while electricity emissions fell by more than half, from 48.1 to 22.3 Mt, as five coal plants closed or switched to natural gas.

Since 2017, every tonne of Alberta's decline came from electricity. Oil and gas kept growing. Alberta also produces more than half of all emissions reported by Canada's large facilities: 152.7 of 291.7 Mt in 2024 (52.4%).

One trap sits in the data. Reported emissions jumped 18.0 Mt in 2017, but about half of that jump came from smaller facilities that only began reporting when the threshold dropped from 50 kt to 10 kt. This brief compares years using only facilities of 50 kt or more, which reported under the same rules throughout.

Key findings

Alberta facilities emitting 50 kt CO2e or more, unless noted.

Measure	2009	2017	2024	Change, 2009 to 2024
All large facilities (Mt)	118.2	149.3	141.6	+23.4 (+20%)
Oil, gas and mining (Mt)	50.1	78.5	91.3	+41.2 (+82%)
Utilities, mainly electricity (Mt)	48.1	45.9	22.3	−25.8 (−54%)
Manufacturing (Mt)	16.5	19.6	21.1	+4.6
Pipelines (Mt)	2.9	4.5	5.9	+3.0

Emissions are also concentrated. In 2024, 220 of Alberta's 769 reporting facilities (29%) produced 93% of its reported emissions, so cutting emissions means dealing with about 200 sites, not 800.

What changed in electricity

Five former coal plants account for the drop. Together they emitted 38.44 Mt in 2017 and 10.11 Mt in 2024, a fall of 74%. Company announcements confirm what happened at each one.

Plant	2017 (Mt)	2024 (Mt)	Change	What happened
Sundance	12.72	0.93	−93%	Most units retired; one switched to gas
Genesee	9.71	4.73	−51%	All units on gas from June 2024
Keephills	8.30	2.72	−67%	One unit retired; two switched to gas
Sheerness	5.66	1.11	−80%	Switched to gas
Battle River	2.05	0.63	−69%	Switched off coal

Three new natural gas plants, led by Cascade near Edson, added 2.79 Mt. Genesee burned coal for about five months of 2024, so its emissions should fall further in 2025. Sheerness and Battle River fell by more than fuel switching alone explains (their owner expected a 45% to 50% cut in emissions per unit of power), which suggests they also generated less. [Inference: the data does not report electricity output.]

National context

Alberta's coal phase-out stands out. In 2024, Saskatchewan's three coal plants (Poplar River, Boundary Dam and Shand) still emitted 7.4 Mt, about 27% of that province's reported facility emissions, and the province plans to keep coal running past the federal 2030 deadline. Coal plants in New Brunswick (Belledune, 2.1 Mt) and Nova Scotia (Lingan, 1.7 Mt) are set to convert or retire between 2028 and 2030.

Data and method

Data. Environment and Climate Change Canada's Greenhouse Gas Reporting Program: emissions by gas for every facility emitting 10 kt CO2e or more, 2004 to 2024 (20,685 facility-years). In 2024 these facilities covered 44% of Canada's emissions. CO2 equivalents use IPCC AR5 values for every year.

Tools and checks. SQL (SQLite) for cleaning and analysis, Power BI for a dashboard, QGIS for a map. Checks before analysis confirmed that facility counts and totals match ECCC's published figures (1,879 facilities and 291.7 Mt in 2024), that every total equals the sum of its gases, and that sector shares rebuilt from industry codes match ECCC's (43%, 29%, 20%). Every number in the dashboard and map was checked against the SQL results.

Comparing years. The reporting threshold fell to 50 kt in 2009 and to 10 kt in 2017, tripling the number of reporting facilities. Trends use facilities of 50 kt or more from 2009, which reported under the same rules throughout.

Limitations

- **Large facilities only.** Vehicles, buildings, most farms and smaller sites are excluded, so this describes industrial emissions, not Alberta's total.
- **Direct emissions only**, close to corporate Scope 1. No purchased electricity or supply chain.
- **Coal's replacement is partly invisible.** The data reports no electricity output, and wind and solar farms emit nothing, so it cannot show how much coal power was replaced by renewables, gas or imports.
- **Missing locations.** In 2024, 16 pipeline and distribution systems (3.4% of emissions) had no single location. They are left off the map but included in every total.

Sources

- ECCC, GHGRP Emissions Data by Facility, data current as of October 2, 2025; Overview of 2024 reported emissions
- Alberta plants: TransAlta, Dec 2021; Heartland Generation, Nov 2021; Capital Power, June 2024; Cascade Power Project, Aug 2020; Power Engineering (Milner M2), Oct 2023
- Other provinces: CBC News, Jan 2025 (Saskatchewan); SaskPower (Shand); CP24, Sept 2026 (Belledune); Global Energy Monitor (Lingan; tracker wiki)